

International Conference on Machine Learning and Data Engineering

MFBFST: Building a stable ensemble learning model using multivariate filter-based feature selection technique for detection of suspicious URL

Sanjukta Mohanty^a, Arup Abhinna Acharya^a^a*School of Computer Engineering, KIIT Deemed to be University, Bhubaneswar, 751024, India*

Abstract

The overwhelming growth and popularity of web motivated the cyber attackers to develop fraudulent web sites and execute numerous attacking strategies to trick the user revealing sensitive information, installing malware automatically through drive-by-download attack, stealing the identity and money etc. Most of the attacking strategies are spreading through compromised URL (Uniform Resource Locator) and it greatly influences the internet performance. Blacklisting approach is adopted to mitigate these issues, but the demerits of this approach is unable to identify the zero-day attacking patterns and is computationally expensive as it has to search the URLs from a large pool of database. So, to plan an effective detection framework for identifying suspicious web sites is a tedious task. To overcome these issues, we have proposed a suspicious URL detection technique for selecting the most influential significant features for classifying the URL as safe or malignant with the help of multivariate filter-based feature selection technique (MFBFST) of machine learning. The redundant features are eliminated by correlation feature selection technique and the significance of relevant attributes are tested with statistical t-test to obtain the most significant features that has more impact on the prediction of malicious websites. The relevant features obtained from MFBFST are used for evaluating the machine learning algorithms like Bagging, Adaboost, GBoost and kNN (k Nearest Neighbour) to create an efficient model for predicting the malicious web sites efficiently. To assess the effectiveness of MFBFST in the enhancement of prediction of classification results we have evaluated the classifier with and without considering the FST also tested the scalability issues by considering two publicly available datasets. Our implementation results demonstrated that with utilizing the CFS, the machine learning algorithms accomplished the highest classification accuracy of 97% in dataset I and 99.25% in dataset II. We have also compared our proposed approach with existing studies in terms of classification accuracy and found our proposed approach achieved significant improvement in detecting the malignant URL.

© 2023 The Authors. Published by Elsevier B.V.

This is an open access article under the CC BY-NC-ND license (<https://creativecommons.org/licenses/by-nc-nd/4.0>)

Peer-review under responsibility of the scientific committee of the International Conference on Machine Learning and Data Engineering

Keywords: Suspicious URL; Ensemble classifier; Multivariate; Feature selection; Machine learning.

1. Introduction

During the past decades the World Wide Web and Internet have become the prevalent platforms for information retrieval, sharing and carrying out the business. The development of sophisticated communication technology coupled with technological advancement encourages the progress and advancement of e-commerce, social networking and e-banking which causes the new opportunities for the invaders. Most of the leading attacking strategies implemented by the attackers are the spreading of compromised URL (Uniform Resource Locator), phishing, XSS (Cross Site Scripting), SQL Injection and drive-by-download attacks to perform the malicious activities. Such attacks incorporate the malicious URL which trick the unsuspecting users to reveal sensitive data such as credit card details, password that leads to financial loss due to theft of identity and also install the malware in user's system [1-3]. So, it is significant to know the components of URL which spreads the malwares. URL refers to a web site or web document or it is the global address of web resources in web. The key aim of the URL is to find the location of the web resources available in the world wide web and provides mechanism for accessing the resources through the web browser. An URL has multiple parts which tells the web browser how and where to retrieve the documents. Different parts of the URL are 1. scheme: identifies the protocol (http, https, gopher, ftp, file etc.) to be used in accessing resources. 2. host name: a unique name represents the complete address to the resource. 3. path: it refers to a file or location in a web server and separated by slash ('/'). 4. Query string: consists of parameter name and value pair separated by delimiter and preceded by question mark ('?'). The structure and components of a URL is represented in Fig. 1.

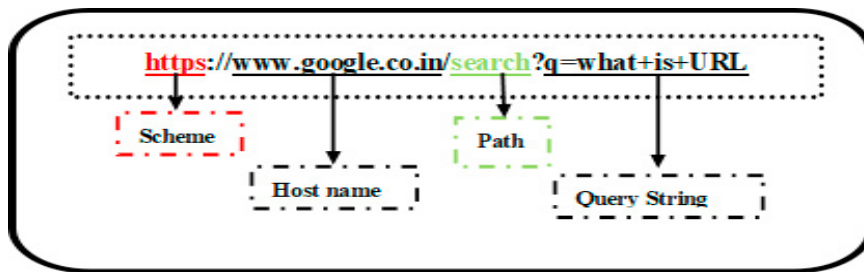


Fig. 1. URL components.

The URLs of the malicious web sites look different from the benign web sites as the length of the later is more than the former, number of dots are also more in malicious web sites. From the URL we can also be able to come to know the more or less reputable hosting center where the web site is hosted as the malignant web sites are hosted in less reputable hosting center. So, from the host name we can infer the hosting center of malignant web sites, who is the owner and their management. The TTL (Time to Leave) value for DNS records related to the domain name helps in finding the malicious web sites as they appear for a short time in comparison to safe web sites [5]. For detecting such malicious websites different approaches have been developed. Black-listing approaches are simple, only identify the known and suspected threats, but unable to identify the threats which are not listed previously in the database. As the new threats are emerging constantly so this method is unable to identify the zero-day attacks. The traditional batch machine learning approach without implementing any effective feature selection technique [2], [3], [5] requires more resources and training time. So, the limitations of black listing approach and more resource intensive look up of traditional machine learning approach motivated us to evolve a new framework for the detection of fraudulent web site.

In this scenario, our research aims at

- To use ensemble machine learning classifier (EMLC) that acts as a distinct paradigm having the capability of joining multiple diverse weak learners to build a model with high predictive power for URL classification (malignant or safe) which provides highest accuracy in comparison to batch machine learning classifiers.
- To present several groups of highly distinctive features which enable our methodology to deliver a substantial and superior performance of identifying the suspicious URL.
- To improve stability and predictive performance we have used MFBFST to choose the relevant and redundant free attributes for reducing the dimensionality and training time.
- The proposed MFBFST approach has adopted some statistical analysis technique like t-test to find the significance association between features. The significance of relevance of features are tested with t-distribution of statistics.
- To check the scalability issues, two numbers of publicly available URL datasets with different dimensionalities are used.
- The results of experiments demonstrate the performance of EMLC enhances after implementing the efficient feature selection technique.
- The efficacy of our approach is achieved by comparing different existing studies with the MFBFST approach for classifying the URL as malicious or benign. The experiment conducted able to evaluate the machine learning classifier accurately hence increases the reliability to some extent.

Though the proposed approach achieves the best performance in classifying the malicious URL still it has some shortcomings.

- We have not tested our approach in real time system where huge amount of data flow occurs in every second.
- Any feature optimization technique has not explored for obtaining the most influential optimal features that has more impact on the prediction of suspicious URL.
- The proposed approach has considered three objectives as maximizing the accuracy of the classification, minimizing the feature dimension and checking scalability issues.
- Still the complexity and computational time are the objectives to be obtained in malicious URL detection.

The rest of the research work is designed as: Section 2 defined the related work. Section 3 explains the background information of different feature selection techniques and bagging and boosting estimators. The proposed approach is depicted in section 4. Experiments and results are explained in section 5. In section 6 we conclude our paper with list of few limitations of the approach proposed and future scope.

2. Related work

A comparative review is conducted on the prediction of malignant web sites using filter-based feature selection methodologies with machine learning techniques. We address some merits and demerits of existing studies for detection of malicious URL and the Table 1 represents the detail comparison of different feature selection techniques for identification of fraudulent web sites.

Table 1. Details of the existing approaches for malicious URL detection.

References	Approach	Brief Description	Remarks
Khonji et al. [6]	Filter and wrapper	Classified the anti phishing email using feature selection techniques.	Effectively found the relevant features . F1 score is 99.3%
Qabajeh et al. [7]	IG, CFS and Chi-Square	12 significant and most relevant features selected to train the classifiers. To investigate the correlation amongst the attributes, number of rules are used within the algorithms.	Used Filtering process affect the classification accuracy. No statistical analysis has been done.
Thabtah et al. [8]	CFS, IG, Chi-Square	To increase the detection rate, from the original 30 features, 3 clusters of features are formed with cut off points by three feature assessment methods. DT algorithm generated the consistent result on same feature's clusters.	Produce significant features in multiple application domain. Proved the goodness or feature significance by using different cutoff points
Sundararajan et al [20]	IG, Chi-square, PCC	Out of 31 extracted features 23 best features are selected using the three separate filtering methods.	The relevant features yields an F1 score from 0.88 to 0.99 FPR =0.01 and FNR=0.00
Shabudin et al [21]	FSOR, FSFM	20 best features were selected from 30 number of features. FSFM is implemented using IG.	The experimental result was not determined by any statistical methodology. Accuracy of RF is 97% after feature selection.
Patil et al [10]	IG, Chi-square, CFS, PCA, LSA, Wrapper	Evaluated 6 feature selection techniques and 8 machine learning classifiers and got minimum performance overhead.	Achieved accuracy of 94-99%, FPR of 0.006-0.096 and FNR of 0.000-0.013. Statistical verification of the relevance of features are not done.
Basnet et al [22]	CFS, wrapper	genetic search and greedy search methods of wrapper methods are used to select the features for evaluation of classifier.	FPR is of 0.2 and FNR is of 0.5
Chen et al [23]	ANOVA, XGBoost	17 features selected out of 48 features after the implementation of ANOVA and XGBoost feature selection function and feature importance ranking.	Accuracy of XGBoost classifier is 99%.

Khonji et al. [6] evaluated a number of feature subset selections method for classifying the phishing email. Filter and wrapper feature selection evaluators were used to find the most effective relevant feature subset from 47 features and evaluate the email classifier accurately with f1 score is of 99.3%. Qabajeh et al. [7] investigated phishing email detection approach considering filter feature selection technique to segregate the high significant features from low significant features and utilized the classifier One Rule, Decision Tree (J45), JRip and part to derive the detection performance. classifiers are derived by using 10 fold cross validation method. The classifiers were tested two times. First with the complete data set consists of 47 features and secondly with 12 minimal numbers of features after applying the different filtering methods. The experimental results revealed that this approach achieved highly competitive result after going through the filtering process. Statistical analysis has not been conducted to determine the significance or goodness of the features. The classification algorithm part (combination of J48 and JRip) had outperformed to One Rule, J48, JRip for classifying the phishing email. Thabtah et al. [8] derived correlation among the common features. The authors identified phishing activities on websites by determining the minimal set of features through comparing the different feature assessment techniques. The filtering methods like CFS, IG, Chi-Square techniques were used against the data set consists of 30 features. Three possible feature clusters were formed by three features assessment

methods using different cut off points to distinguish between significant and insignificant features. The data mining classification algorithm C4.5, PART, RIPPER was utilized to measure the accuracy of filtering methods on different features. Sundararajan et al [20] focused on assessing phishing and benign web pages based on F1 score, false positive rate by integrating filter and wrapper feature selection methods with machine learning classifiers RF, LR (Logistic Regression) and neural network (multilayer perceptron). Gathered data from phish tank and majestic million. Out of 31 extracted features, 23 features are selected as best from mutual information statistical test which greatly impact the predictive performance to increase from 0.88 to 0.99 (F1 score) also false positive rate was 0.01. Shabudin et al [21] considered two FSM such as Feature Selection by Omitting Redundant feature and Feature Selection by Filter Method using classification accuracy. Three machine learning algorithms RF, multilayer perceptron and Naive Bayes were used for evaluating the performance of phishing web pages. The experimental result gave that the RF classifier with FSFM feature selection technique produced highest performance of 97.08% of accuracy. An effective methodology was proposed by Patil et al [10] to identify the malignant web pages from the dataset consists of 117 features from both static and dynamic web pages and 52,082 samples of malignant and safe web pages. Six feature selection techniques were adopted like chi-square, IG (Information Gain), correlation feature selection (CFS), Principal Component Analysis (PCA), LSA (Latent Semantic Analysis) and wrapper method for selecting the relevant features. Out of 117 features 15 relevant features were selected for training the eight machine learning estimators like J48 decision tree (DT), Logistic Regression (LR), Adaboost, Random Tree (RT), SGD (Stochastic gradient descent), BayesNet, REPTree, Majority Voting. Majority voting estimator with wrapper feature selection technique achieved highest accuracy of 99.15%. Then the authors verified their results with some software tools like anti-virus and anti-malware but lacking the statistical verification of the significance of features which may leads to increase the classification performance. CFS and WFS methods were used to choose significant attributes in Basnet et al [22] to improve the phishing detection. Out of considered 177 features of page content and URL based features, 42 relevant features are selected through correlation based genetic search and 12 features by greedy forward search of wrapper method. The classifiers NB, LR, RF and MLP are used to detect the usefulness of the selected relevant attributes. Though wrapper approach worked in comparison to CFS but was extremely slow. Chen et al [23] proposed a stable URL classification model to detect the malicious URL efficiently. ANOVA and XGBoost feature selection functions are adopted to minimize the dimension of the attributes. XGBoost estimator is trained by applying 10-fold cross validation technique.. Along with the XGBoost, KNN and SVM (support vector machine) classifiers are evaluated with feature selection functions for prediction of suspicious web sites. XGBoost classifier achieved highest accuracy of 99%.

The pitfalls of existing approaches are lack of statistical analysis in feature-feature association. To our best knowledge though many of the feature selection methodologies are adopted for obtaining the relevant features for malicious URL detection but significant association has not been determined by any of the approaches. Hence to fill this research gap, a secure detection framework has provided where after removing the correlated features the significance of independent features are tested with class level features to filter out the most relevant features that evaluates the boosting estimators for predicting the URL as malignant and benign because the boosting estimators are very successful learning models to enhance the performance intensively.

3. Methodologies used

3.1. Feature selection technique (FST)

In data mining, one of the most important pre-processing job is feature selection whose key objective is to find a small and best subset of pertinent attributes from the data set that reduces the redundancy and maximizes the relevance to target class in classification task [15]. Feature selection can be mathematically expressed as: Given a dataset ‘D’ contains ‘n’ number of features represented as $D = \{X_1, X_2, \dots, X_n\}$. The objective is to select the best subset from D according to some evaluation criterion. Extract subset $s = \{X_1, X_2, \dots, X_s\}$ where $s < n$ and $X_1, X_2, \dots, X_n$ represents the attributes of any dataset. FSM comprises of four fundamental phase, namely, formation of subsets (subsets generated for evaluation basing upon certain search strategy), evaluation of subsets (evaluating and comparing with the preceding best feature according certain evaluation criteria), stopping criterion and result validation (best subset

validated with validation set) [11,15,16]. To obtain the most relevant and redundant free best features, several feature selection techniques are developed. Generally, FSM are categorized into two types: filter selection technique and wrapper selection technique [1,13,4,12]. In filter selection approach, irrelevant features are removed basing upon some criteria like distance, consistency, dependency, information and correlation (general characteristics of training data) and some statistical measures or intrinsic properties are used to select the goodness of features without involving the machine learning classifiers [14]. The filter methods are again categorized in to two groups: univariate and multivariate. In univariate group, each feature is ranked or measured independently according to some evaluation criterion and the dependency or relationship between the features are not considered. The univariate filter techniques are such as F-test, chi-square, information gain and t-statistics etc. The multivariate scheme tries to find the relationship between two attributes and the techniques included are Correlation based Feature Selection (CFS), Mutual Information Feature Selection (MIFS), Analysis of Variance (ANOVA), minimum Redundancy Maximum Relevance (mRMR), Fast Correlation Based Feature Selection (FCBF), Max-Relevance-Max-Distance (MRMD), etc. Multivariate filter-based feature selection methods are though slower than the univariate methods but more efficient because it removes the redundancy features. In contrast to filter methods, wrapper approaches are depended on learning classifier to find relevant feature subsets hence incurs high computational cost in large data set but leads to more accuracy. In this paper we have considered the multivariate filter-based feature selection technique for finding the most relevant features because these methods are independent of any learning classifier, trust on general characteristics of training sample, lower computational cost and good generalization ability and the CFS method finds the features which are more correlated with or predictive of the class [10].

3.2. Ensemble machine learning technique

In ensemble machine learning technique several base weak algorithms are integrated together to construct a stable strong learner for better prediction performance rather than using a single classifier which may give lower metric accuracy. Ensemble learning approach solves the issues of under fitting and over fitting of traditional machine learning classifier which occurs due to noise, variance and bias in data sets. The reason for using ensemble classifier is it's improved predictive performance, generation of a stable model and the ability to remove the class imbalance. Generally, the ensemble learning methods are categorized into two types: Bagging method and Boosting method.

In bagging methods, the data set is divided randomly and the base estimator is constructed randomly on each of these random sampling. Base estimators are executed in parallel with replacement. The prediction results of individual classifier are aggregated to obtain the final prediction. The final estimator is represented in equation (1) as follows:

Where Fe denotes final estimator. $Be_1, Be_2 \dots Be_n$ represents base estimators.

$$Fe = \sum_{i=1}^n (Be_i) / n \quad (1)$$

Bagging ensemble approach is used to reduce the bias [24]. Bias is calculated as the average of difference between predicted and actual results.

In Boosted methods base learners are built sequentially and assign weights like $w_1, w_2, \dots, w_n$ to each sample of training set. Initial weight is set to $w_i = 1/N$ than it is modified in successive iteration. The succeeding model tries to correct the error produced in the previous model. The procedure is repeated till it reaches the terminating condition (correctly predicting the whole dataset or maximum number of models). Boosting approach is used to reduce the variance [24].

4. Proposed method

This section explained a proposed methodology based on MFBFST for detecting suspicious URL. The proposed approach conducted through 4 stages like collection of data sets (consists of both legitimate and benign websites), preprocessing phase, learning or training phase, evaluation phase and forecasting phase which is depicted in Fig. 2.

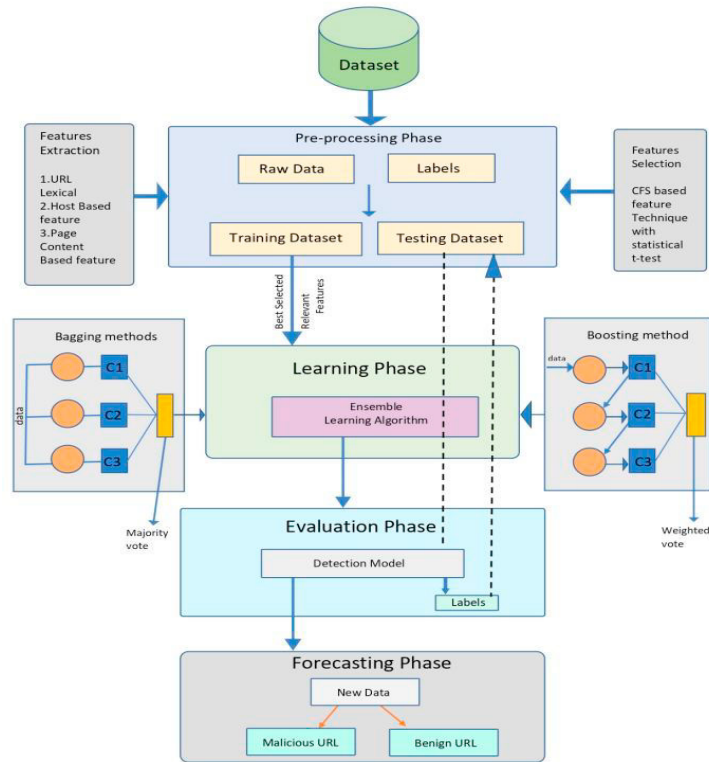


Fig. 2. Schematic sketch of proposed approach for malicious URL prediction.

4.1. Collection of datasets

To assess the performance of the different ensemble classifiers for the prediction of suspicious web sites, two numbers of publicly available datasets are considered from a data science community, UCI machine learning repository and Kaggle databases [17][18]. The characteristics of the dataset is shown in Table 2.

Table 2. Different characteristics of datasets.

Name	Features	Instances	Classes
DS-I	31	11,055	2
DS-II	50	10,000	2

4.2. Preprocessing phase

In this section, data preparation tasks are carried out for extracting the meaningful information required for training the learning classifiers for effectively identifying the malicious web sites.

4.2.1. Feature selection technique

In this section, different literatures have been reviewed to identify the significant most popular website-based features which are employed in the field of malicious URL detection. Extracting the discriminative website's features to train the machine learning classifier to increase the accuracy of detection is a tedious task. From the literature we have identified there are three categories of features presented for malicious URL detection: URL based lexical feature, host-based feature and page content-based feature. URL based Lexical features include the structural properties of the URL which refers to length of URL, components of URL (domain features, directory related features, file name features), number of special characters etc. The justification of considering lexical feature is: we can learn a classifier in a shorter time as no need to download the page for the classification. Simply we have to pay attention on the structural characteristics of the URL because malicious URL looks different from the safe URL. Host based features includes hosting center of malicious web sites, who is the owner of that malicious site, its registration date, expiry date, how these malicious sites are managed and TTL (Time to Live) value of DNS (Domain Name Server) record associated with host name etc. This feature is included in the proposed data set because unlike the benign web sites the malicious websites are hosted in a less reputable hosting center and they appear in a shorter time interval. The page content-based features are the html and JavaScript based feature. Though the extraction of these content-based features takes more time but it contains more information about a web site hence it leads to better prediction model. In this research study we have considered all these three web site features for identification of malignant web sites.

4.2.2. CFS and significance test

After the preparation of discriminative websites features, CFS is applied to obtain the most essential and relevant features to assess the ensemble learning estimators for enhancing the malicious web sites detection. The CFS technique aims to find the features which are highly correlated with the target class and uncorrelated with each other [11]. Correlation between independent feature and class feature represents how accurately an independent feature can predict a class of malicious or benign. It evaluates and ranks the feature in the search space of all possible features using the correlation (a data intrinsic properties) based heuristic evaluation function [25] represented in equation (2) as:

$$r = \frac{n(\sum xy) - (\sum x)(\sum y)}{\sqrt{[n(\sum x^2) - (\sum x)^2][n(\sum y^2) - (\sum y)^2]}} \quad (2)$$

Where 'r' specifies the correlation coefficient, 'x' represents independent feature, 'y' represents the target feature and 'n' species the no of observations. The value of r varies from +1 to -1. +1 species two variables are strongly correlated, -1 says the variables are negatively correlated and 0 means two variables have no correlation. CFS has been evaluated on two training datasets and relevant features are selected to learn the machine learning classifiers. Generally, the CFS method is used to obtain the relation or association between the features for a data sample. But if we consider for the whole data set, the correlation coefficient may not produce the goodness of result. Hence, we have used t-test of correlation coefficient to test whether the relation between dependent and independent feature is statistically significant. We have implemented this t-test on correlation coefficient to choose the most significant attributes from the dataset. The appropriate t-value is computed from t-distribution [25] that is represented in equation (4) with the assumption that there is no correlation between the independent feature and target variable ($\rho=0$) and the alternate hypothesis $\rho=1$ represents that there is a relationship between independent features and class label feature.

$$t = r \sqrt{\frac{n-2}{1-r^2}} \quad (3)$$

Where ‘n’ shows the number of observations, ‘r’ denotes the correlation coefficient for sample data points.

The significance of the relationship is expressed in probability form (significant at $\alpha=0.05$). If the t-value of a feature is more than the critical value at $t=0.05$, significant level then this feature is statistically significant and it is selected for training the classifiers to identify the malicious URL which is explained in the Algorithm 1.

Algorithm 1: Finding statistically significant features for malicious URL detection from the URL dataset based on t-distribution.

Input: A sample of training dataset: $D = \{f_1, f_2, \dots, f_n, f_c\}$ where $f_1, f_2, \dots, f_n$ represents the independent features and f_c denotes the class label feature.

Output: D_{best} (relevant and statistically significant feature)

Begin

$\rho=0$ represents no relation between two variables.

$\alpha=0.05$ (significant label)

$n-2$ shows degree of freedom.

CV states the critical value at significant level 0.05 when degree of freedom is $n-2$ (calculated from the correlation coefficient table [19])

(2) for each feature $j=1$ to n do

(3) r = calculate correlation coefficient with (f_j, f_c)

(4) $t = r \sqrt{\frac{n-2}{1-r^2}}$

(5) If $t > \text{critical value (CV)}$

set $\rho=1$ (relation exists between independent and class variables)

(6) $D_{\text{best}} = j$

(7) End

(8) return D_{best}

(9) End

4.3. Learning phase

After selecting the significant features from feature selection technique, the ensemble learning classifier are learned with the training sample to design a model that can be able to correctly classify the malicious URL. Then Bagging techniques of bagging methods and Adaboost and Gradient Boost classifiers of boosting methods are used for predicting the malignant URL. Also, traditional machine learning classifier kNN (k Nearest Neighbour) algorithm is considered to compare the performance of the ensemble estimators with batch machine learning algorithm. The detail description of bagging and boosting methods are explained in section 3.2.

4.4. Evaluation phase

To find the best, effective and feasible model, the model evaluation phase is an important process to represent the data. The learned classifiers in the learning phase are tested with new test sample to get the different performance measures of classification precisely. The model for malignant URL detection is evaluated by confusion matrix, a performance measure or a metric which contains different elements like True Positive (TP), True Negative (TN), False Positive (FP) and False Negative (FN) [9]. Confusion matrix is represented in table 3. We have considered malicious link as negative example and assign class label ‘1’ to it and begin web site as positive examples and set ‘0’ to it. TP represents the total number of malicious URL that are detected correctly. TN denotes the total number of benign URL that are identified correctly. FP shows the total number of malicious URL which are tested as benign. FN results the total number of benign URL identified as malicious.

Table 3. Elements of confusion matrix.

Actual \ Predicted	Malicious URL (1)	Benign URL (0)
Malicious URL (1)	True Positive (TP)	False Negative (FN)
Benign URL (0)	False Positive (FP)	True Negative (TN)

4.5. Model Performance Metrics:

$$Accuracy = \frac{TP + TN}{TP + FP + TN + FN} \quad (4)$$

$$Precision = \frac{TP}{TP + FP} \quad (5)$$

$$Recall = \frac{TP}{TP + FN} \quad (6)$$

$$F1-score = \frac{2 * precision * recall}{precision + recall} \quad (7)$$

5. Results of Experiments

The proposed model (MFBFST) is evaluated effectively for the classification of the malignant URL. Also, the statistical t-test is used for finding the significance of relevant features which are generated from CFS feature selection technique are effective to classify the web sites correctly and whether the ensemble model is fit to classify the websites as safe or risky. To resolve the scaling issues of the proposed approach, two numbers of malicious URL data sets are collected from internet sources, UCI machine learning repository [17][18]. The experiment is carried out by jupyter notebook, an editor of python environment for executing and evaluating the approach and matplotlib a data visualization package of python for plotting the graphs. We have considered some commonly used measures, which are described, in section 4.4 to measure the performance and observe the effectiveness of the ensemble classifier. The performance of machine learning classifier KNN, Bagging, Adaboost and GBoost are represented in Table 4.

Table 4. Performance comparison of different machine learning algorithms without and with considering the feature selection techniques using two numbers of data sets.

Performance measure before applying selection technique					Performance measure after applying selection technique				
DS-I					DS-II				
Classifiers	Accuracy	Precision	Recall	F1-Score	Classifiers	Accuracy	Precision	Recall	F1-Score
Bagging	0.99	1.00	0.95	0.97	Bagging	0.97	1.00	0.92	0.96
Adaboost	0.93	0.83	1.00	0.91	Adaboost	0.96	1.00	0.89	0.89
GBoost	0.95	0.86	0.95	0.90	GBoost	0.95	0.89	0.96	0.92
KNN	0.99	1.00	0.95	0.97	KNN	0.97	1.00	0.81	0.89
DS-I					DS-II				
Classifiers	Accuracy	Precision	Recall	F1-Score	Classifiers	Accuracy	Precision	Recall	F1-Score
Bagging	0.93	0.91	0.92	0.91	Bagging	0.99	1.00	0.98	0.99
Adaboost	0.95	0.92	0.92	0.92	Adaboost	0.98	1.00	0.99	0.99
GBoost	0.97	0.96	0.96	0.96	GBoost	0.99	1.00	0.96	0.96
KNN	0.97	0.96	0.97	0.97	KNN	0.99	1.00	0.97	0.95

Then CFS, a multivariate filter method for selecting the relevant features is used. The parameter, thresh hold value (d) of 0.8 is considered for adding or removing a feature in CFS. If the value of correlation coefficient (r) of any

feature is greater than the value of d than that feature is added otherwise it is removed and it is determined from the correlation matrix. Due to the limited space, we have not shown the correlation matrix in this paper. To confirm validity of the selected features generated from CFS methods, again a statistical t-test is performed to check the significance of relevant features. For our experiment we have considered the parameters as number of observations as 5 and degree of freedom as 3 and the value of significance level α is 0.05. So, when the degree of freedom is 3, the critical value (CV) at significance level 0.05 is calculated from the correlation coefficient table [24] as 0.878339. Then these values of the α and CV are applied in the proposed Algorithm 1, to find the best significant features for enhancing the learning process to detect the web sites as malignant and benign. When the CFS method is implemented in the DS-I, three numbers of feature are found to be highly correlated with each other hence these features are dropped from the dataset because they may have negative impact on the classification results and select the rest numbers of 23 features out of 31 features for statistical t-test for testing the significance of each feature. Similarly, in DS-II, 6 features were found as highly correlated and after dropping these correlated features we got 43 numbers of features out of 50 features as relevant for predicting the class as malicious and benign. The features, chosen from CFS are again tested using statistical t-test for finding the significance between independent feature and class level feature.

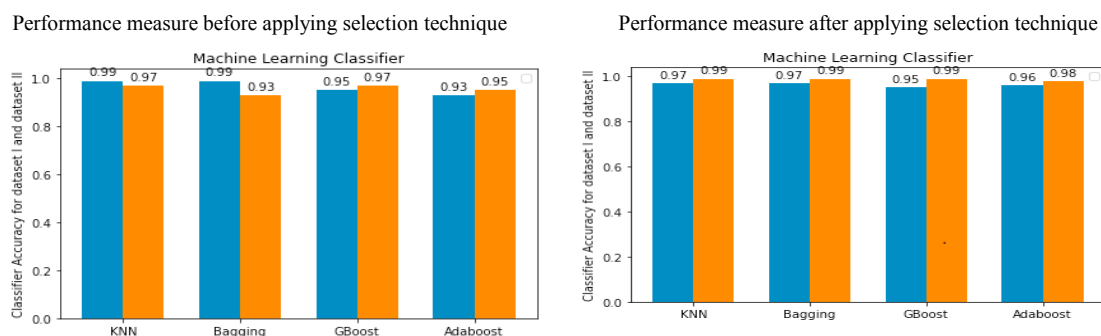


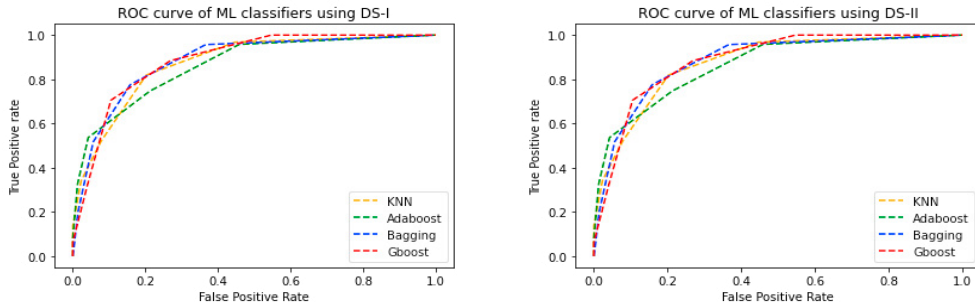
Fig. 3. Accuracy of different ensemble learning estimators before and after implementing the feature selection techniques

Experimental Evaluation of statistical t-test on Dataset I & II

We have calculated the t-test value of independent feature 'URL_Length' and class level feature 'Result' by implementing the Algorithm 1. as 2.347655 which is greater than critical value (CV=0.878339) which shows there exist a relationship between features 'URL_Length' and 'Result'. Hence the alternate hypothesis $\rho=1$ is accepted and the null hypothesis $\rho=0$ is rejected and the independent feature 'URL_Length' is statistically significant and relevant feature for training the classifiers to detect the new URL as malignant and benign. In this manner, we have tested for all the features of DS-II also. After obtaining the relevant features from feature selection techniques the classifiers are evaluated and recorded the overall accuracy, precision, recall and F1-score for detecting the malicious websites.

The detection accuracy obtained by different machine learning classifiers after the FST are represented in Table 5 and in Fig. 3. It is evident from the Fig. 3, after the deployment of CFS technique, the accuracy of different classifiers converges and stabilize to a stable state of accuracy that is the maximum accuracy level remain constant even the increased dimension of features. When we compared the accuracy among the machine learning classifiers, all most all the classifiers except Adaboost estimator outperformed consistently with a score of 99% in DS-II and above 95% in DS-I. So our proposed approach can be able to handle the large dataset using MFBFST in fraudulent website detection. Then the quality of the proposed approach is validated by ROC (Receiver operative characteristics) graph improved by CFS technique which is represented in Fig. 4. We have plotted ROC graph for different classifiers on both the datasets with and without implementing FST. It can be noticed from the Fig. 4. the classification model Gboost produces a point in the upper left corner of coordinate (0,1) depicting the true positive rate value as almost '1' which specifies that the model perfectly distinguish the malicious class from benign class.

Performance of classifiers before feature selection



Performance of classifiers after feature selection

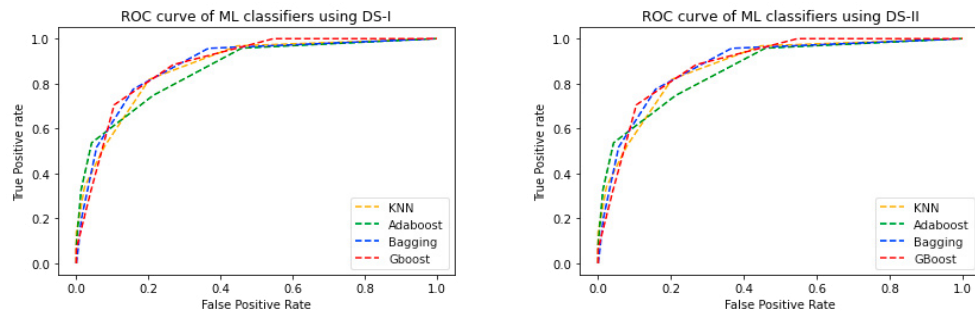


Fig. 4. ROC curve of different classifiers with and without considering the feature selection technique in DS-I and DS-II.

To validate the proposed approach with the existing studies [21], [22], [20], [23], [10], [26] we have compared the proposed approach in terms of classification accuracy with existing methods for the identification of malicious URL and it has been depicted in the Fig. 5. After deploying the CFS with statistical t-test the performance of the classifiers of the proposed approach increases compared with other approaches that implement only CFS, IG and Chi-square feature selection techniques. In research study [26] though the authors used boosting ensemble estimators still achieved the classification accuracy as 98.66%. It can be observed from the Fig. 5. The proposed model accomplished the best and highest performance in terms of accuracy of classification for suspicious URL identification. The TPR of the proposed model for the classification of malicious URL is approximately 1 which is represented in Fig. 4 in comparison to other research studies where TPR is low and FPR is 0.2 in [22].

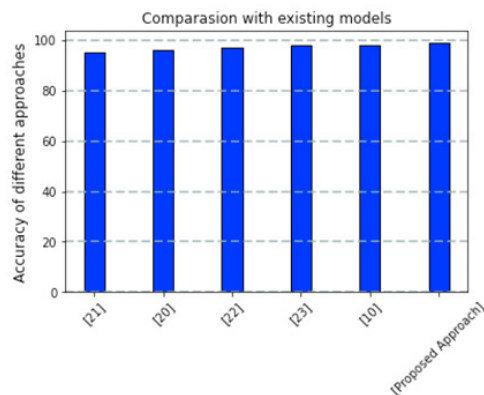


Fig. 5. Comparison of our proposed method with existing methods considering classification accuracy.

6. Conclusion

The internet security is affected by the criminal activities of intruder through malicious web sites. Hence to keep away the users from browsing those malignant URLs, a suspicious web sites detection framework is essential which is a challenging task now a days as the malicious content of the web page is changed from time to time. We have suggested a suspicious URL detection method for selecting the most influential significant features for classifying the URL as safe or malignant with the help filter based multivariate feature selection technique of machine learning. The redundant features are eliminated by CFS and the significance of relevant features are tested with statistical t-test to obtain the most significant features that has more impact on the prediction of malicious websites. The relevant features obtained from FST are used for evaluating the machine learning algorithms Bagging, Adaboost and GBoost and KNN to create an efficient model for predicting the malicious web sites efficiently. To assess the effectiveness of FST in the enhancement of prediction of classification results we have evaluated the classifier with and without considering the FST also tested the scalability issues by considering two datasets. Our implementation results demonstrated that with utilizing the CFS, the machine learning classifier achieved the highest classification accuracy of 97% in dataset I and 99.25% in dataset II. We have also compared our proposed approach with existing methods in terms of classification accuracy and found our proposed approach achieved significant improvement in detecting the malignant URL. Thus, a future model can be designed by using the advanced network features which can be able to improve the detection rate in real time system. To enhance the prediction of suspicious websites we will adopt some feature optimization techniques to select the significant features with less the computational time.

References

- [1] Sahoo D, Liu C, and Hoi S C H. (2017) "Malicious URL detection using machine learning: A survey." arXiv preprint arXiv:1701.07179.
- [2] Catak F, O Sahinbas, K, Dörtkardeş V. (2021) "Malicious URL Detection Using Machine Learning." *In Artificial Intelligence Paradigms for Smart Cyber-Physical Systems, IGI Global*: pp. 160-180.
- [3] Kumar H, Gupta P, Mahapatra R P. (2018) "Protocol based Ensemble Classifier for Malicious URL Detection." *In: 3rd International Conference on Contemporary Computing and Informatics, ICCCI. IEEE*: pp. 331-336.
- [4] Gomez A L P, Maimo F, Clemente F J G. (2019) "A Deep Learning-based System for Network Cyber Threat Detection." *Machine Learning for Computer and Cyber Security*, pp. 1-25, CRC Press.
- [5] MaJ, Saul L, Savage S Voelker G M. (2009) "Beyond blacklists: learning to detect malicious web sites from suspicious URLs." *In: Proceedings of the 15th ACM SIGKDD International conference on Knowledge discovery and data mining*, pp. 1245-1254.
- [6] Khonji M, Jones A, Iraqi Y. (2013) "An empirical evaluation for feature selection methods in phishing email classification." *International Journal of Computer Systems Science & Engineering*, **28**(1): 37-51.
- [7] Qabajeh I, Thabtah F. (2014) "An experimental study for assessing email classification attributes using feature selection methods." *In 3rd International Conference on Advanced Computer Science Applications and Technologies, ICACST.IEEE*, pp. 125-132.
- [8] Thabtah F, Abdelhamid N. (2016) "Deriving correlated sets of website features for phishing detection: a computational intelligence approach." *Journal of Information & Knowledge Management*, **15**(04): 1650042
- [9] Li T, Kou G, Peng Y. (2020) "Improving malicious URLs detection via feature engineering: Linear and nonlinear space transformation methods." *Information Systems*, **91**, 101494.
- [10] Patil D R, Patil J B. (2019) "Malicious web pages detection using feature selection techniques and machine learning." *International Journal of High-Performance Computing and Networking*, **14**(4): 473-488.
- [11] Ali W. (2017) "Phishing website detection based on supervised machine learning with wrapper features selection." *International Journal of Advanced Computer Science and Applications*, **8**(9): 72-78.
- [12] Chiew K L, Tan C L, Wong K, Yong K S, Tiong W K. (2019) "A new hybrid ensemble feature selection framework for machine learning-based phishing detection system." *Information Sciences*, **484**, 153-166.
- [13] Ali W, Malebary S. (2020) "Particle swarm optimization-based feature weighting for improving intelligent phishing website detection." *IEEE Access*, **8**, 116766-116780.
- [14] Ali W, Ahmed A A. (2019) "Hybrid intelligent phishing website prediction using deep neural networks with genetic algorithm-based feature selection and weighting." *IET Information Security*, **13**(6): 659-669.
- [15] Tang J, Alelyani S, & Liu H. (2014) "Feature selection for classification: A review." *Data classification: Algorithms and applications*, 37.
- [16] Liu H, Yu L. (2005) "Toward integrating feature selection algorithms for classification and clustering." *IEEE Transactions on knowledge and data engineering*, **17**(4): 491-502.
- [17] Dua D and Graff C. (2021) UCI Machine Learning Repository. School of Information and Computer Science, University of California, Irvine, CA, USA. <https://archive.ics.uci.edu/ml/datasets/Phishing+Websites>. Accessed 20 June 2021.

- [18] Tan, Choon Lin (2018) “Phishing Dataset for Machine Learning: Feature Evaluation”, Mendeley Data, V1, Universitiy Malaysia Sarawak Faculty of Computer Science and Information Technology, <https://www.kaggle.com/shashwatwork/phishing-dataset-for-machine-learning>, <https://doi.org/10.17632/h3cgnj8hft.1>, Accessed 10 May 2021.
- [19] G Stephanie (2022) PPMC Critical Values From StatisticsHowTo.com: Elementary Statistics for the rest of us!. 2022, <https://www.statisticshowto.com/tables/ppmc-critical-values/> Accessed Feb 05 2022.
- [20] Sundararajan A, Gressel G, Achuthan K. (2019) “Feature Selection for Phishing Detection with Machine Learning.” *International Journal of Engineering and Advanced Technology (IJEAT)*, Vol-8 Issue-6S3, Blue Eyes Intelligence Engineering & Sciences Publication, pp.1039-1045 <https://doi.org/10.35940/ijeat.F1331.0986S319>.
- [21] Shabudin S, Sani N S, Ariffin K A Z, Aliff M. (2020) “Feature selection for phishing website classification.” *International Journal of Advanced Computer Science and Applications*. 11(4): 587-595, <https://doi.org/10.14569/IJACSA.2020.0110477>.
- [22] Basnet R. B, Sung A. H, & Liu Q. (2012) “Feature selection for improved phishing detection.” *In International Conference on Industrial, Engineering and Other Applications of Applied Intelligent Systems*, pp. 252-261, Springer, Berlin, Heidelberg.
- [23] Chen Y C, Ma Y W, Chen J L. (2021) “Intelligent Malicious URL Detection with Feature Analysis.” *In 2021 IEEE Symposium on Computers and Communications, ISCC.IEEE*, pp. 1-5.
- [24] Singh P, Ranga V. (2021) “Attack and intrusion detection in cloud computing using an ensemble learning approach.” *International Journal of Information Technology*, **13(2)**: 565-571.
- [25] Blessie E C, Karthikeyan E. (2012) “Sigmis: a feature selection algorithm using correlation-based method.” *Journal of Algorithms & Computational Technology*, **6(3)**: 385-394.
- [26] Mohanty S, Achary A. A , Sahu L. (2022).” Improving Suspicious URL Detection through Ensemble Machine Learning Techniques.” *In Society 5.0 and the Future of Emerging Computational Technologies*, Book chapter no-13 (pp. 229-248). CRC Press.